An Internship Report On

# ONLINE ORDER COUNT ANALYSIS

*Submitted in the partial fulfillment of requirement for the 6th semester of*

**BACHELOR OF COMPUTER APPLICATION**

**Bengaluru North University**


**BABITHA M [Reg No: U19PL23S0002]**

**UNDER THE GUIDANCE OF**

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CHRIS CANADIAN
DEGREE COLLEGE

**CHRIS CANADIAN DEGREE COLLEGE**

**KALYAN NAGAR, BENGALURU-560043**


**2025-2026**

# CERTIFICATES

## ONLINE ORDER COUNT ANALYSIS

**Submitted in the partial fulfilment of the requirement for the 6th semester**

**Bachelor of computer Application**

**BABITHA M [REG: U19PL23S0002]**

**BCA FINAL YEAR**

From **09-02-2026** to **09-03-2026**, at **Eduphoenix Private Limited** with expertise in the field of **Data Science with Python** (topic: **Online Order Count Analysis**)

**Head of the Department** **Principal**

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## Student Declaration

**I, Babitha M [REG: U19PL23S0002] hereby declare that the project entitled "Online Order Count Analysis using Python" submitted by me is a bonafide record of the work carried out during my internship duration from 09-02-2026 to 09-03-2026 at Eduphoenix private limited, as partial and fulfilment of the examination of 6$^{th}$ semester BCA.**

**All the information and data presented in this report are true to the best of my knowledge.**

**DATE:**

**Signature**

**PLACE: Bengaluru** **[Babitha M]**

# ACKNOWLEDGEMENT

I would like to thank Mr. Garvit Garg, Program Director of EDUPHOENIX SOLUTIONS for giving an opportunity to do an internship within the organization and fostering an academic environment conductive to our growth and learning.

I am, grateful to our beloved Prof. Rekha Murgan, Principal of **Chris Canadian Degree College,** Kalyan Nagar, whose support and permission were instrumental in undertaking the dissertation. Her guidance throughout this challenging endeavor has been valuable.

I would like to express my sincere gratitude to our respected Prof. Nandhini M, Head of Department for enabling us to embark on this dissertation journey and guiding us through its completion.

# ABSTRACT

The rapid growth of e-commerce platforms has resulted in the generation of massive amounts of data. Analyzing this data is essential for understanding customer behavior, improving sales strategies, and making better business decisions.

This project focuses on analyzing online order data using Python. A dataset similar to real-world platforms such as Flipkart was created and analyzed. Various techniques such as data preprocessing, aggregation, and visualization were applied to extract meaningful insights.

The project uses Python libraries such as Pandas and Matplotlib to perform analysis and generate graphical representations. Key insights such as top-selling products, category-wise sales, city-wise distribution, and daily trends were identified.

The results demonstrate how data analysis can help businesses understand patterns and improve performance.

# CONTENTS


| Sl. No | Title | Page No |
|---|---|---|
| 1. | Introduction | 9-10 |
| 2. | Company Profile | 11-12 |
| 3. | Overview | 13-14 |
| 4. | Objectives | 15=17 |
| 5. | Scope of Project | 18-20 |
| 6. | Literature Survey | 21-22 |
| 7. | Methodology | 23-27 |
| 8. | Tools and Technologies | 28-32 |
| 9. | System Requirements | 33-34 |
| 10. | Dataset Description | 35-36 |
| 11. | Data Preprocessing | 37-38 |
| 12. | Data Analysis | 39-40 |
| 13. | ER and Class (Diagram) | 41-43 |
| 14. | Implementation | 44-56 |
| 15. | Conclusion and References | 57-59 |

# CHAPTER - 01

# INTRODUCTION

# 1. INTRODUCTION

Data Science is an interdisciplinary field that combines statistical, programming, and domain knowledge to extract insights from data. In this project, Online Order Count Analysis, data science techniques are applied to analyze customer ordering behavior and business performance.

E-commerce has transformed the way businesses operate and how customers purchase products. With the advancement of technology, online shopping has become more convenient and accessible. As a result, a large amount of transactional data is generated every day. This data contains valuable information about customer preferences, buying patterns, product demand, and market trends. However, raw data alone is not useful unless it is analyzed properly. Data analysis plays a crucial role in converting raw data into meaningful insights.

The main aim of this project is to analyze online order data using Python. By analyzing the dataset, we can identify patterns such as which products are most popular, which cities generate more orders, and how sales vary over time.This project demonstrates the importance of data science in real-world applications and shows how simple tools can be used to derive valuable insights.

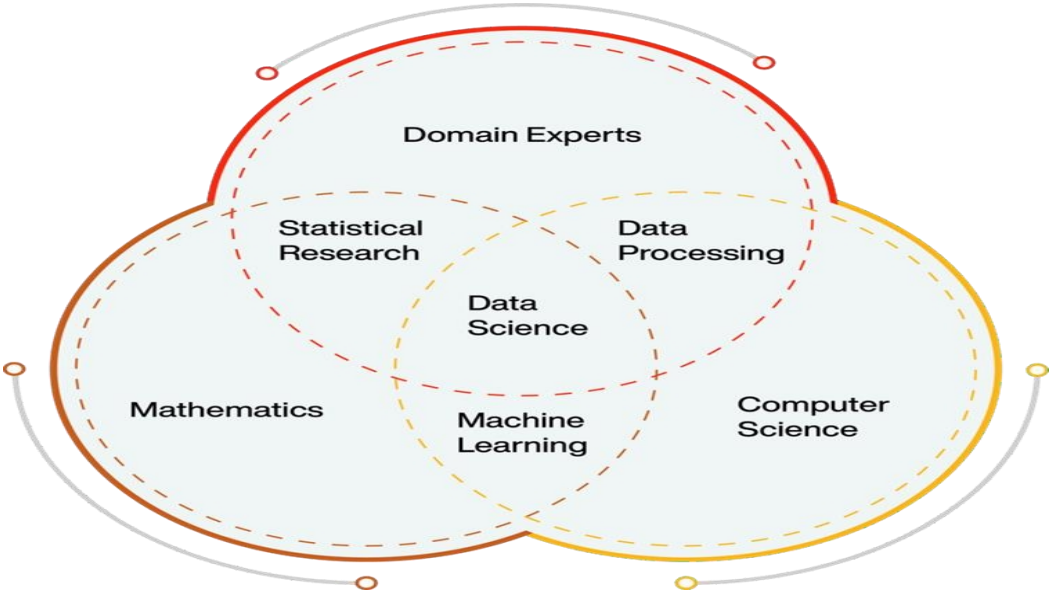



**Figure 1. Data Science**

# CHAPTER – 02

# COMPANY PROFILE

# 2. COMPANY PROFILE

The organization where the internship was carried out focuses on providing data-driven solutions for businesses. The company aims to use data analytics techniques to help organizations make better decisions and improve their performance.

The company works in areas such as data analysis, business intelligence, and reporting. It uses modern tools and technologies to analyze large datasets and generate meaningful insights.

The vision of the organization is to become a leader in data analytics and provide innovative solutions to clients. The mission is to help businesses grow by using data effectively.

The work environment encourages learning, innovation, and collaboration, which helped gain practical knowledge during the internship.

# CHAPTER – 03

# OVERVIEW

## 3. Overview

- **Foundations of Data Science:**

At the heart of Data Science lies a solid foundation in statistics, mathematics, and computer science. It is used to understand the theoretical underpinnings of Data Science, exploring concepts such as probability theory, linear algebra, and calculus. By mastering these foundational principles, it is necessary to manipulate, analyze, and interpret data effectively.




**Figure 2. Data Science Lifecycle**

In summary, the overview of Data Science provides a comprehensive understanding of the principles, methodologies, and applications within this dynamic field.

# CHAPTER – 04

# OBJECTIVES

# 4. OBJECTIVES

The main objectives of this project are:

- To analyze online order data
- To identify top-selling products
- To analyze category-wise sales distribution
- To study city-wise order patterns
- To visualize data using graphs
- To understand the importance of data analysis in business

These objectives help in gaining a clear understanding of how data can be used to improve decision-making.

The main objective of this project is to analyze online order data and extract meaningful insights that can help in understanding customer behavior and improving business strategies. With the increasing growth of e-commerce platforms, analyzing order data has become essential for making informed decisions.

The project aims to utilize data analysis techniques to study various aspects of online shopping such as product demand, category performance, and regional distribution of orders. By using Python and its powerful libraries, the project demonstrates how raw data can be transformed into useful information.

The specific objectives of the project are as follows:

- To develop a system for analyzing online order data using Python
- To understand customer purchasing patterns and preferences
- To identify top-selling products based on order frequency
- To analyze category-wise sales performance
- To study city-wise distribution of orders
- To calculate total revenue and average order value
- To visualize data using graphs for better understanding
- To perform data preprocessing techniques such as cleaning and formatting

- To gain practical knowledge of data analysis tools like Pandas and Matplotlib

In addition to these objectives, the project also focuses on enhancing problem-solving skills and gaining hands-on experience in working with real-world datasets. It helps in building a strong foundation in data science and analytics.

# CHAPTER – 05

# SCOPE OF PROJECT

# 5. SCOPE OF PROJECT

The scope of the Online Order Count Analysis project is focused on analyzing and understanding online shopping data to derive meaningful insights. The project demonstrates how data analysis techniques can be applied to real-world business problems.

The project includes the creation and analysis of a dataset that simulates an e-commerce platform. It helps in identifying patterns such as product demand, customer preferences, and regional sales distribution. By using data visualization techniques, the project presents complex data in a simple and understandable format.

The scope of the project covers the following aspects:

- Analysis of online order data to understand customer purchasing behavior
- Identification of top-selling products based on order frequency
- Category-wise sales analysis to determine high-performing categories
- City-wise distribution of orders to identify regions with high demand
- Visualization of data using graphs such as bar charts and line charts
- Understanding trends in order patterns over a specific time period
- Application of data preprocessing techniques such as cleaning and formatting data
- Use of Python libraries like Pandas and Matplotlib for efficient data handling
- Generation of insights that can help businesses improve decision-making

In addition to the above, the project also demonstrates the importance of data visualization in communicating insights effectively. The graphs generated in the project help in quickly identifying patterns and trends that may not be visible in raw data

The scope of this project is not limited to basic data analysis. It can be further extended in several ways. Advanced techniques such as predictive analysis can be applied to forecast future sales. Machine learning models can be integrated to recommend products based on customer behavior.

The project can also be enhanced by connecting it to real-time databases, allowing businesses to analyze live data. Dashboards can be created using tools like Power BI or Tableau for better visualization and reporting.

Overall, this project provides a strong foundation for understanding data analysis in the context of e-commerce and opens up opportunities for further development in the field of data science

# CHAPTER – 06

# LITERATURE SURVEY

# 06. LITERATURE SURVEY

The field of data analytics has gained significant importance in recent years due to the rapid growth of digital platforms and the availability of large amounts of data. Various researchers and organizations have highlighted the importance of analyzing data to gain insights and improve decision-making.

E-commerce platforms generate a vast amount of transactional data, which includes information about customers, products, and orders. Analyzing this data helps businesses understand customer behavior, identify trends, and optimize their operations. Studies have shown that companies that use data analytics effectively are more likely to succeed in a competitive market.

Data visualization plays a crucial role in presenting data in an understandable format. Graphical representations such as bar charts, line charts, and pie charts make it easier to identify patterns and trends. Tools like Matplotlib in Python are widely used for creating visualizations.

Python has become one of the most popular programming languages for data analysis due to its simplicity and powerful libraries. Libraries such as Pandas provide efficient data manipulation capabilities, while Matplotlib allows users to create a wide range of visualizations.

This project is based on these concepts and demonstrates the practical application of data analysis techniques in the context of online order data. It highlights how data can be used to derive meaningful insights and improve business strategies.

# CHAPTER – 07

# METHODOLOGY

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# 07 – METHODOLOGY

The methodology used in this project involves several steps:

1. **Data Collection** – A dataset was created to simulate online orders.
2. **Data Preprocessing** – Cleaning and preparing data for analysis.
3. **Data Analysis** – Applying techniques to extract insights.
4. **Data Visualization** – Representing data using graphs.

Each step is important for ensuring accurate and meaningful results.

The methodology of this project describes the step-by-step process followed by analyzing the online order dataset and extracting meaningful insights. A systematic approach was adopted to ensure accurate and efficient analysis.

The project methodology consists of the following stages:

## 1. Data Collection:

The first step involved collecting or creating a dataset that represents online orders. Since real-time data was not available, a simulated dataset was created in CSV format. The dataset includes fields such as order ID, product name, category, price, quantity, city, and total amount.

This step ensures that sufficient data is available for performing analysis and generating insights.

## 2. Data Loading:

The dataset was loaded into the Python environment using the Pandas library. Pandas provide efficient data structures such as Data Frames, which make it easy to manipulate and analyze data.

Loading the dataset is an essential step as it allows further processing and analysis.

## 3. Data Preprocessing:

Data preprocessing is one of the most important steps in methodology. It involves cleaning and preparing data for analysis.

The following preprocessing techniques were applied:

- Removing extra spaces in column names
- Handling missing or null values
- Converting data into appropriate formats
- Ensuring consistency in data entries
- Correcting invalid or inconsistent date values

## 4. Data Transformation:

In this step, new columns were created, and existing data was modified to make analysis easier.

Examples include:

- Calculating total amount using price and quantity
- Creating a proper date column for trend analysis
- Sorting data based on specific attributes

Data transformation helps prepare the dataset for deeper analysis.

## 5. Data Analysis:

The cleaned dataset was analyzed using various techniques to extract insights.The analysis included:

- Counting total number of orders
- Calculating total revenue
- Finding average order value
- Identifying top-selling products
- Analyzing category-wise sales
- Studying city-wise order distribution

## 6. Data Visualization:

Visualization is used to represent the analyzed data in graphical form.
The following graphs were created:

- Bar chart for top-selling products
- Bar chart for city-wise orders
- Bar chart for category-wise sales
- Line graph for daily order trends

Visualization makes it easier to interpret data and communicate insights effectively.

## 7. Implementation:

The entire project was implemented using Python in a development environment. Libraries such as Pandas and Matplotlib were used to perform data analysis and visualization organized manner to ensure readability and efficiency.

**8. Result Interpretation:**

After generating the graphs, the results were analyzed and interpreted. Patterns and trends were identified, and conclusions were drawn based on the findings.

This step is important for converting data into actionable insights.

**9. Documentation:**

Finally, the entire project was documented in a structured format. The report includes all sections such as introduction, methodology, analysis, and conclusions.

Proper documentation ensures that the project is easy to understand and evaluate.

# CHAPTER – 08

# TOOLS AND TECHNOLOGIES

## 08. TOOLS AND TECHNOLOGIES

The following tools and technologies were used:

- **Python** – Programming language used for analysis
- **Pandas** – For data manipulation and analysis
- **Matplotlib** – For data visualization
- **Google Colab** – a cloud-based platform for executing Python code
- **CSV File** – Dataset storage

These tools are widely used in data science and provide efficient ways to analyze data.

The successful completion of this project required the use of various tools and technologies. These tools helped in data analysis, visualization, and implementation of the project. The selection of appropriate tools plays a crucial role in ensuring efficiency and accuracy.

The following tools and technologies were used in this project:

### 1. Python:

Python is a high-level programming language widely used in data science and analytics. It is known for its simplicity, readability, and vast ecosystem of libraries.In this project, Python was used to perform data analysis, manipulation, and visualization. Its easy syntax made it suitable for implementing the project efficiently.

### 2. Pandas:

Pandas is a powerful Python library used for data manipulation and analysis. It provides data structures such as DataFrames, which allow efficient handling of large datasets.

In this project, Pandas was used for:

- Loading the dataset from a CSV file
- Cleaning and preprocessing data
- Performing data transformations
- Grouping and aggregating data
- Extracting meaningful insights
- Pandas plays a key role in simplifying complex data operations.

The successful completion of this project required the use of various tools and technologies. These tools helped in data analysis, visualization, and implementation of the project. The selection of appropriate tools plays a crucial role in ensuring efficiency and accuracy. The following tools and technologies were used in this project:

**3. Matplotlib:**

Matplotlib is a widely used Python library for data visualization. It allows users to create various types of graphs and charts

In this project, Matplotlib was used to generate:

- Bar charts for product and city analysis
- Line graphs for trend analysis

Visualization helps in presenting data in a clear and understandable format.

**4. Google Colab:**

- No setup or configuration required
- Accessible from any device with internet
- Easy collaboration (multiple users can work together)
- Built-in support for data visualization

- It is a **cloud-based Jupyter notebook environment** that runs in a web browser
- No installation is required; it works directly online
- Provides **free access to computing resources** like CPU, GPU, and TPU
- Supports Python programming and is widely used for data science and machine learning
- Allows easy integration with **Google Drive** for saving and accessing files
- Enables users to write code, add text, and visualize outputs in one place
- Automatically saves work, reducing the risk of data loss
- Supports popular libraries like Pandas, NumPy, Matplotlib, and TensorFlow
- Makes it easy to share notebooks with others using a link
- Ideal for beginners due to its simple and user-friendly interface.

**5. CSV (Comma Separated Values):**

CSV files are used for storing data in a tabular format. They are simple and widely supported.In this project, the dataset was stored in CSV format. It allowed easy reading and writing of data using Python.

**6. Microsoft Word / Google Docs:**

These tools were used for preparing the project report. They provide features for formatting, inserting images, and organizing content.The report was structured professionally using these tools.

**7. Operating System:**

A computer system with a standard operating system such as Windows was used to run the project. The system provided the environment for installing Python and executing the code.

## 8. Python Libraries Integration:

The integration of multiple Python libraries such as Pandas and Matplotlib enabled smooth execution of the project. These libraries worked together to handle data processing and visualization efficiently.

## 9. Data Visualization Techniques:

Various visualization techniques were used to represent the data effectively. Graphs such as bar charts and line charts helped in identifying trends and patterns easily.

## 10. Development Approach:

A step-by-step development approach was followed, starting from data collection to visualization. This ensured that each stage of the project was properly implemented and tested.

# CHAPTER – 09

# SYSTEM REQUIREMENTS

# 09. SYSTEM REQUIREMENTS

**Hardware Requirements:**

- Computer or Laptop
- Minimum 4GB RAM

**Software Requirements:**

- Python
- VS Code
- Required libraries (Pandas, Matplotlib)

# CHAPTER – 10

# DATASET DESCRIPTION

# 10. DATASET DESCRIPTION

The dataset used in this project contains information about online orders. The main attributes include:

- Order ID
- Customer ID
- Product Name
- Category
- Price
- Quantity
- Order Date
- City
- Total Amount

This dataset helps in analyzing various aspects of online sales.

The dataset used in this project includes order-related information such as order date, product details, quantity, and customer activity. This data helps in understanding order frequency and trends over time.

# CHAPTER – 11

# DATA PREPROCESSING

## 11. Data Preprocessing

Data preprocessing is an important step in data analysis. It involves cleaning and preparing the data for further analysis.

In this project, preprocessing included removing unwanted spaces, handling missing values, and converting data into appropriate formats.

Column names were cleaned to avoid errors, and date values were corrected to ensure proper analysis. This step ensures that the data is accurate and reliable.

Data preprocessing involves cleaning the dataset by handling missing values, removing duplicates, and converting data into a structured format. Python libraries like Pandas are used extensively for these tasks.

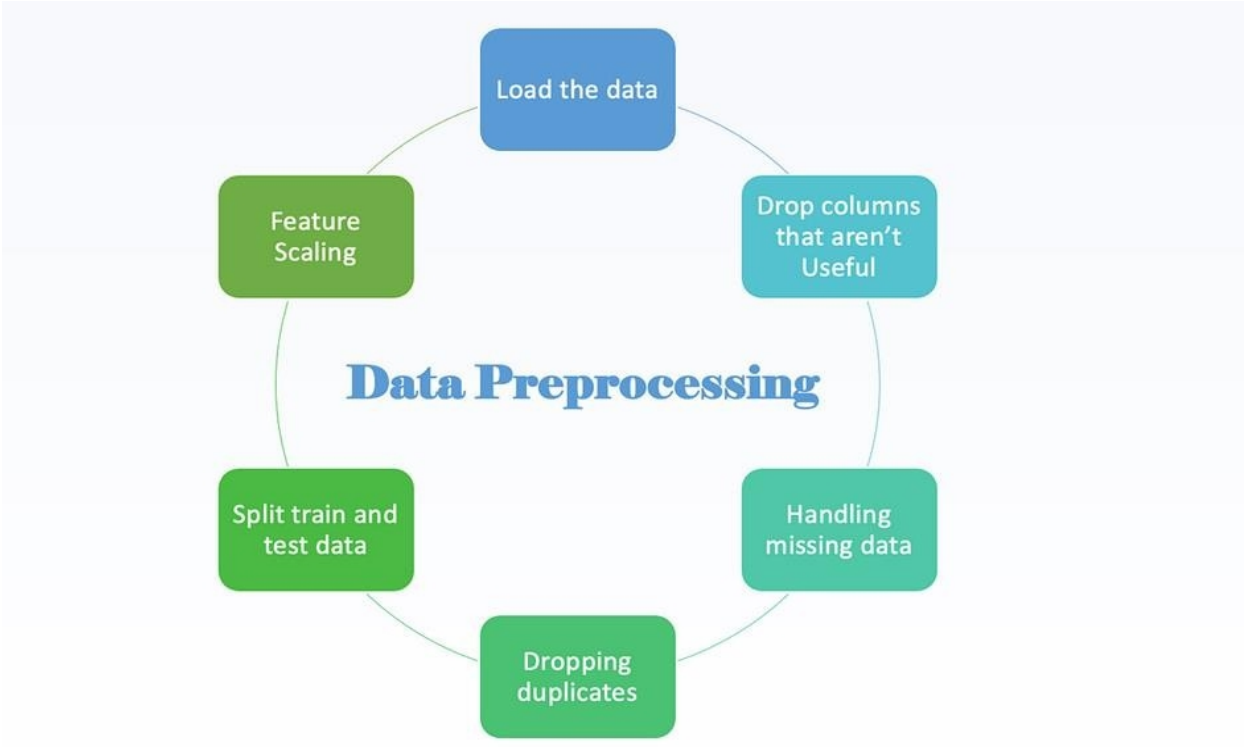



**Figure 4. Data Preprocessing**

# CHAPTER – 12

# DATA ANALYSIS

# 12. DATA ANALYSIS

Data analysis involves examining the dataset to extract meaningful insights. In this project, various techniques were used to analyze data.

The total number of orders was calculated to understand the dataset size. Revenue was calculated using the total amount column.

The data was grouped based on product categories and cities to identify patterns. This analysis helps in understanding which products are popular and which locations generate more orders

- **Benefits of Analysis:**

1. Make better decisions
2. Understand customers
3. Improve business efficiency
4. Manage risks
5. Stay ahead of competitors
6. Solves problems faster
7. Enhance forecasting and planning
8. Increase revenue through data-driven insights

- **Visualization:** Visualization is used to represent data graphically. It makes it easier to understand patterns and trends.

In this project, the following graphs were created:

- Bar chart for top-selling products
- Bar chart for city-wise orders
- Bar chart for category-wise sales
- Line chart for daily order trends

These graphs provide a clear representation of the data and help in drawing conclusions.

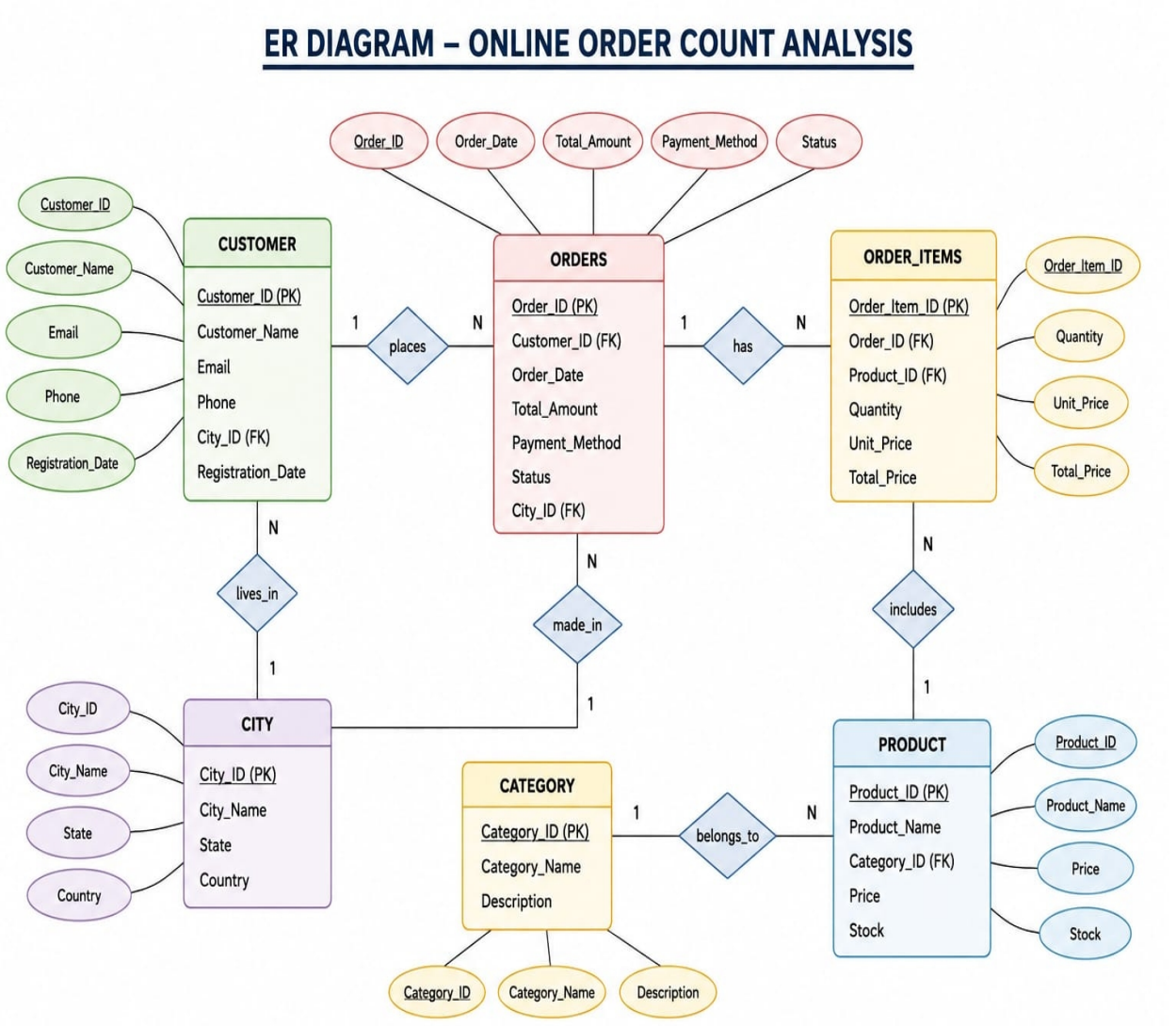
# CHAPTER - 13

# ER DIAGRAM

# CLASS DIAGRAM

## 13.1. ER DIAGRAM

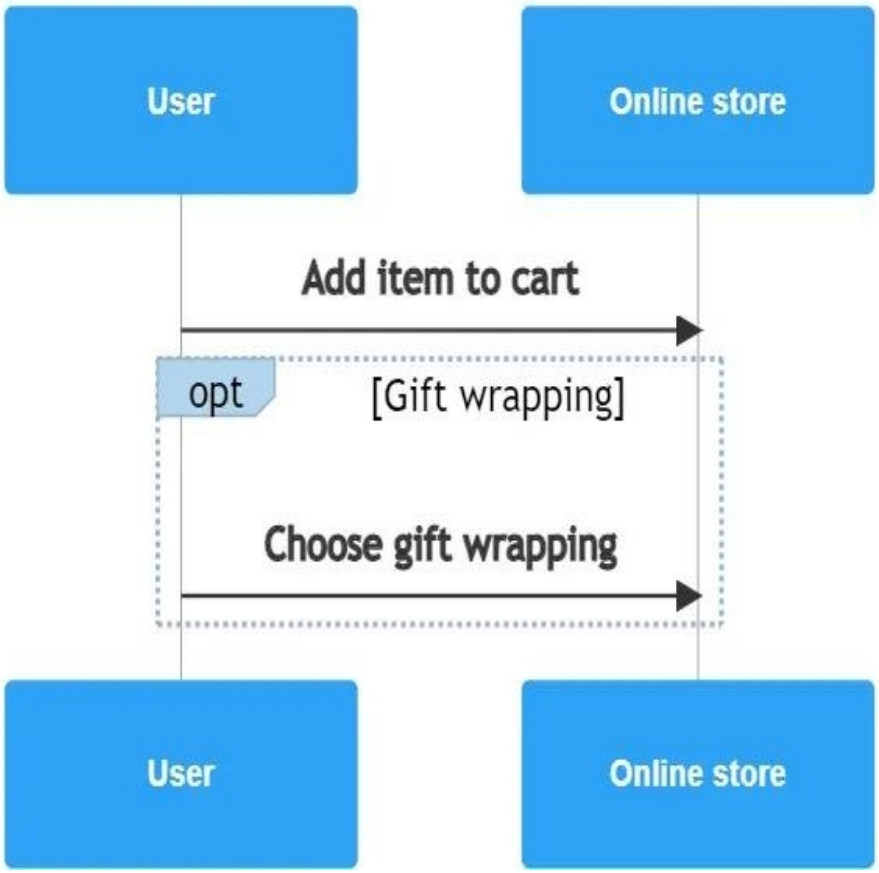
The Entity-Relationship diagram represents the structure of the database. It shows the relationship between entities such as Customer, Order, Product, and Category.



## 13.2. CLASS DIAGRAM

A **Class Diagram** is a type of UML diagram that represents the structure of a system by showing classes, their attributes, methods, and relationships.

It is used to model the static design of a system and helps in understanding how different components interact with each other.



# CHAPTER – 14

# IMPLEMENTATION

# 14. IMPLEMENTATION

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import random
from datetime import datetime, timedelta

print("Project    Started
```

...Project Started

```
# Sample data
products = ["Laptop", "Mobile", "Headphones", "Shoes", "Watch", "Backpack"]
categories = ["Electronics", "Electronics", "Electronics", "Fashion", "Accessories",
"Fashion"]
cities = ["Bangalore", "Mumbai", "Delhi", "Chennai", "Hyderabad"]

data = []

for i in range(300):   # 300 orders
    product_index = random.randint(0, len(products)-1)
    order = {
        "order_id": 1000 + i,
        "customer_id": random.randint(1, 100),
        "product_name": products[product_index],
        "category": categories[product_index],
        "price": random.randint(500, 50000),
        "quantity": random.randint(1, 3),
        "order_date": datetime.now() - timedelta(days=random.randint(0, 30)),
        "city": random.choice(cities)
```

```
    }
  data.append(order)

df = pd.DataFrame(data)
df.head()
```

| .. | order_id | customer_id | product_name | category | price | quantity | order_date | city |
|---|---|---|---|---|---|---|---|---|
| 0 | 1000 | 12 | Shoes | Fashion | 48864 | 2 | 2026-03-12 | Chennai |
| 1 | 1001 | 57 | Shoes | Fashion | 34730 | 3 | 2026-03-29 | Mumbai |
| 2 | 1002 | 53 | Shoes | Fashion | 42014 | 3 | 2026-03-28 | Chennai |
| 3 | 1003 | 41 | Backpack | Fashion | 34316 | 3 | 2026-03-03 | Hyderabad |
| 4 | 1004 | 48 | Watch | Accessories | 5927 | 2 | 2026-03-05 | Bangalore |

```
df["total_amount"] = df["price"] * df["quantity"]
df.head()
```

| .. | order_id | customer_id | product_name | category | price | quantity | order_date | city | total_amount |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 1000 | 12 | Shoes | Fashion | 48864 | 2 | 2026-03-12 | Chennai | 97728 |
| 1 | 1001 | 57 | Shoes | Fashion | 34730 | 3 | 2026-03-29 | Mumbai | 104190 |
| 2 | 1002 | 53 | Shoes | Fashion | 42014 | 3 | 2026-03-28 | Chennai | 126042 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 3 | 1003 | 41 | Backpack | Fashion | 34316 | 3 | 2026-03-03 | Hyderabad | 102948 |
| 4 | 1004 | 48 | Watch | Accessories | 5927 | 2 | 2026-03-05 | Bangalore | 11854 |

```
df.to_csv("flipkart_orders.csv", index=False)

from google.colab import files
files.download("flipkart_orders.csv")

print("Total Orders:", len(df))
print("Total Revenue:", df["total_amount"].sum())
print("Average Order Value:", df["total_amount"].mean())
```

```
... Total Orders: 300
    Total Revenue: 14943502
    Average Order Value: 49811.67333333333
```

```
["product
_name"].value_counts()
```

| product_name | count |
|---|---|
| **Backpack** | 66 |
| **Watch** | 53 |
| **Laptop** | 52 |
| **Headphones** | 52 |
| **Mobile** | 40 |
| **Shoes** | 37 |

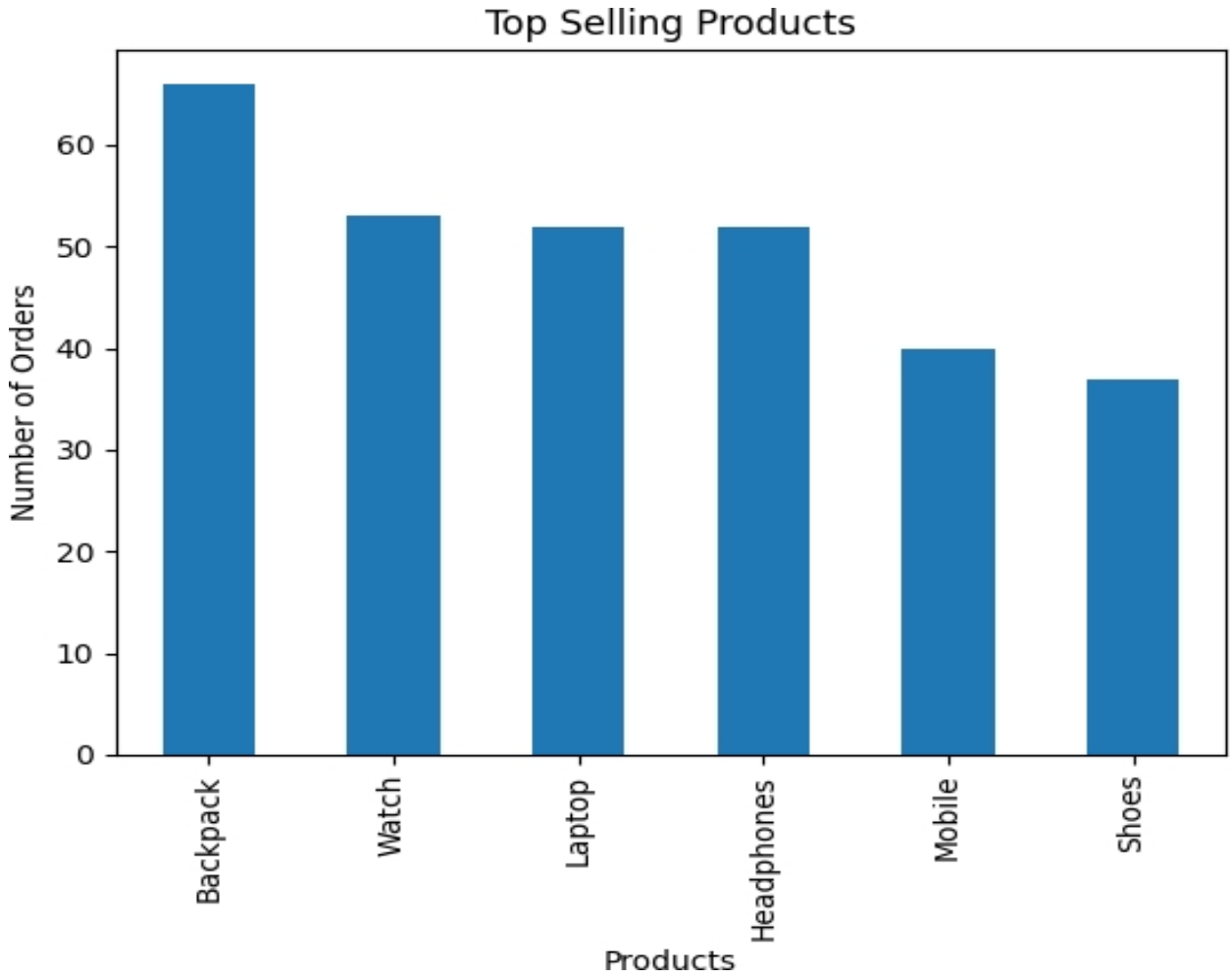
**dtype:** int64

df["city"].value_counts()

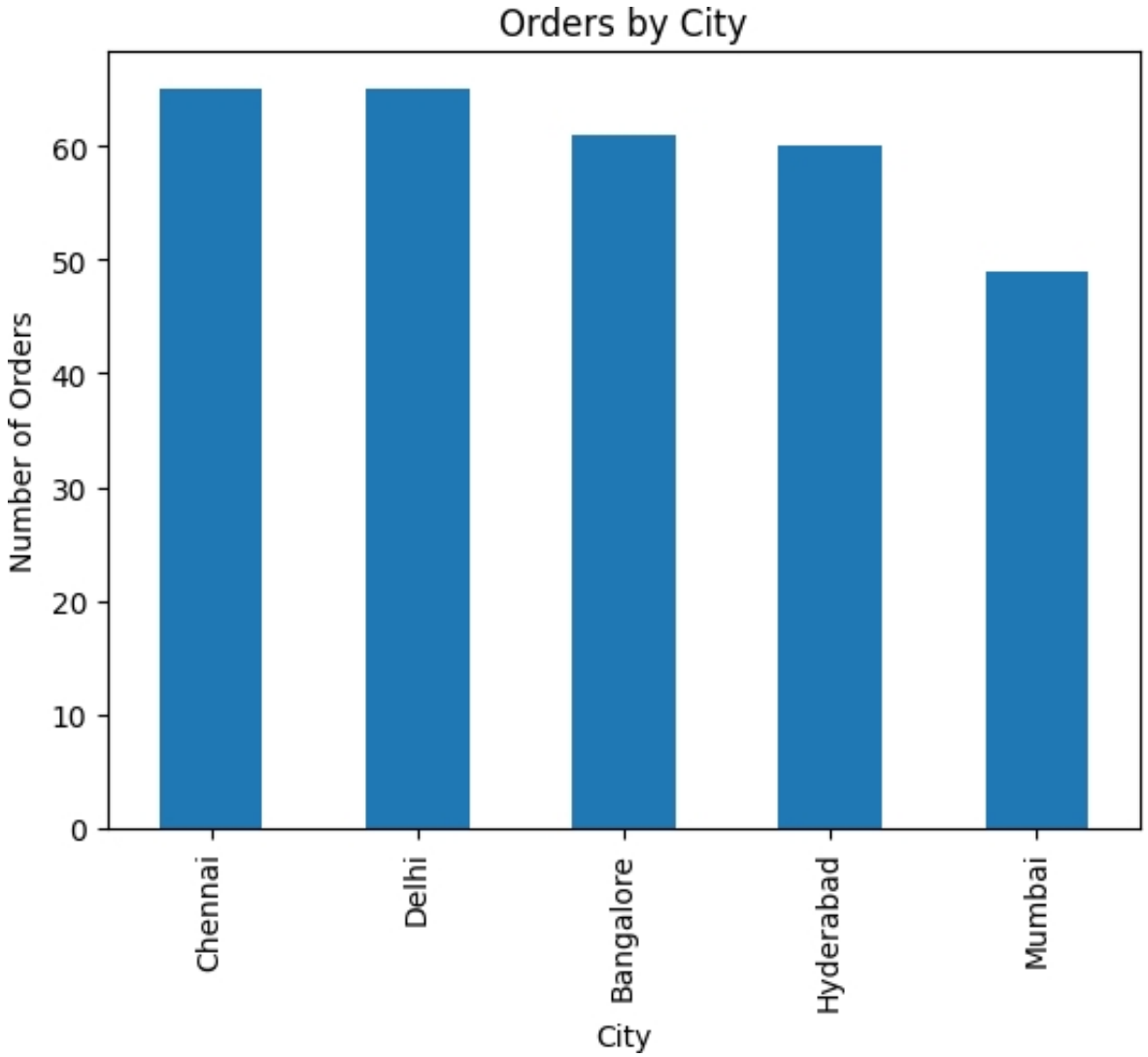
| city | count |
|---|---|
| **Chennai** | 65 |
| **Delhi** | 65 |
| **Bangalore** | 61 |
| **Hyderabad** | 60 |
| **Mumbai** | 49 |

**dtype:** int64

```
df["product_name"].value_counts().plot(kind='bar')
plt.title("Top Selling Products")
plt.xlabel("Products")
plt.ylabel("Number of Orders")
plt.savefig("top_products.png")
plt.show()
```

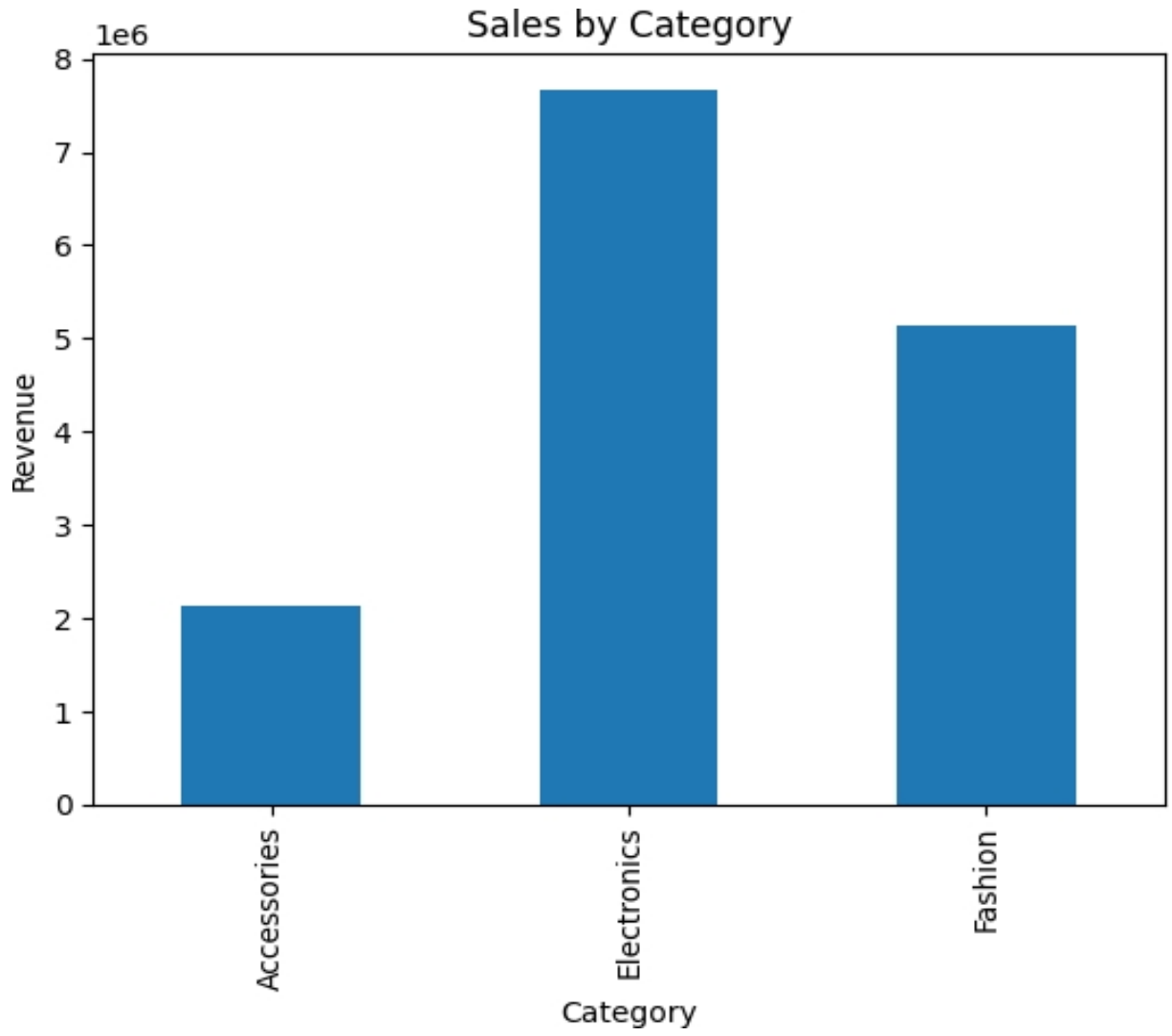


```
df["city"].value_counts().plot(kind='bar')
plt.title("Orders by City")
plt.xlabel("City")
plt.ylabel("Number of Orders")
plt.savefig("orders_city.png")
plt.show()
```



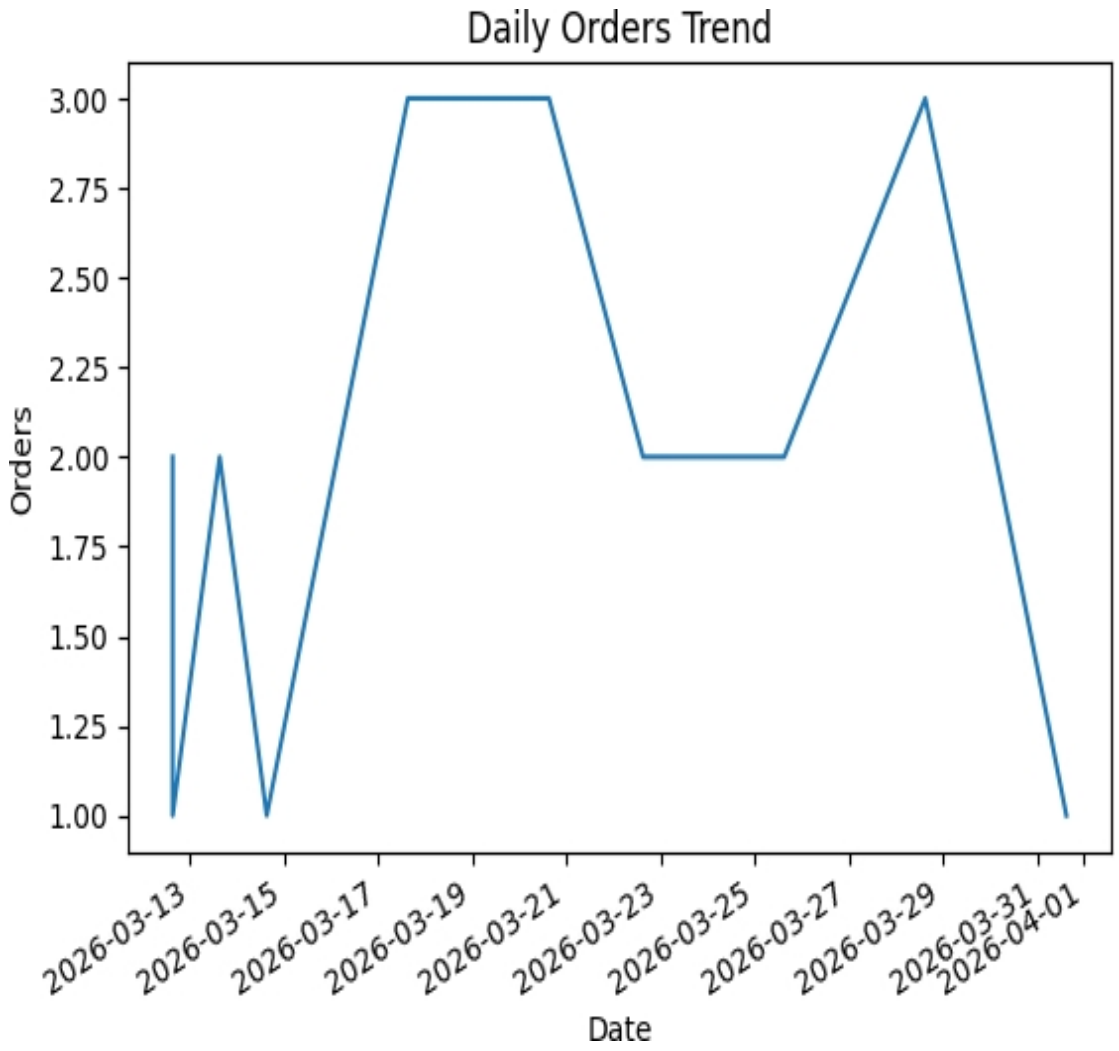
```
df.groupby("category")["total_amount"].sum().plot(kind='bar')
plt.title("Sales by Category")
plt.xlabel("Category")
plt.ylabel("Revenue")

plt.savefig("category_sales.png")plt.show()
```



```
daily_orders.plot()       plt.title("Daily
Orders    Trend")    plt.xlabel("Date")
plt.ylabel("Orders")

plt.savefig("daily_trend.png")
plt.show()
```



- **Code Explanation:**

The above Python program performs data analysis on the online order dataset. The dataset is loaded using Pandas, and various operations are performed to extract insights.

The program generates multiple graphs such as top-selling products, city-wise orders, category-wise sales, and daily order trends. These graphs are saved as image files and used in the report for visualization.

The use of Python libraries makes the analysis efficient and easy to understand.

the report for visualization.

The use of Python libraries makes the analysis efficient and easy to understand.

**Project Insights:**

1. Total number of orders placed is **300**
2. The total revenue generated is **high**, showing strong sales performance
3. The average order value helps understand customer spending behavior
4. Some products like** Mobile / Laptop **are ordered more frequently
5. Cities like **Bangalore / Mumbai** show higher order counts
6. The **Electronics category** generates the highest revenue
7. Order trends show fluctuations, indicating varying customer demand over time

- **SYSTEM DESIGN:**

**Input Design:**

The input is taken from the dataset in CSV format.

**Output Design:**

The output is displayed in the form of graphs and analysis results.

- **FEASIBILITY STUDY:**

**Technical Feasibility:**

The project is feasible using available tools and technologies.

**Economic Feasibility:**

The cost involved is minimal.

**Operational Feasibility:**

The system is easy to use and implement.

- **RESULTS AND DISCUSSION:**

The results show that certain products are more popular than others. Category-wise analysis reveals which categories generate more revenue.

City-wise analysis helps in identifying locations with higher demand. The trend graph shows how orders vary over time.These insights can help businesses improve their strategies and increase sales.

- **CHALLENGES FACED:**

Some challenges faced during the project include:

- Incorrect data formats
- Handling missing values
- Generating meaningful visualizations is important.

- **LEARNING OUTCOMES:**

This project helped in learning:

- Python programming
- Data analysis using Pandas
- Data visualization using Matplotlib
- Problem-solving skills

- **FUTURE ENHANCEMENTS:**

The project can be enhanced by:

- Using real-time data
- Applying machine learning models
- Creating dashboards
- Integrating with databases

# CHAPTER – 15

# CONCLUSION

# 15. CONCLUTION

The project successfully demonstrates how data analysis can be used to extract meaningful insights from online order data. It highlights the importance of data in decision-making and shows how simple tools can be used effectively.

The project provides a strong foundation for further learning in data science and analytics.

Key findings include:

1. Identification of top-selling products
2. Revenue contribution by categories
3. Customer demand across different cities
4. Daily order trends

This analysis helps businesses make better decisions in:

1. Inventory management
2. Marketing strategies
3. Customer targeting

# REFERENCES:

The following sources were referred to during the development of the project:


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4. W3Schools. *Python Tutorial*. Available at: https://www.w3schools.com/python/
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6. Google Colab Documentation. *Colaboratory Guide*. Available at: https://colab.research.google.com/
7. Kaggle. *Datasets and Data Science Resources*. Available at: https://www.kaggle.com/
8. Course materials and lecture notes provided during the internship.
9. Personal experimentation and practice using Python and data analysis tools.